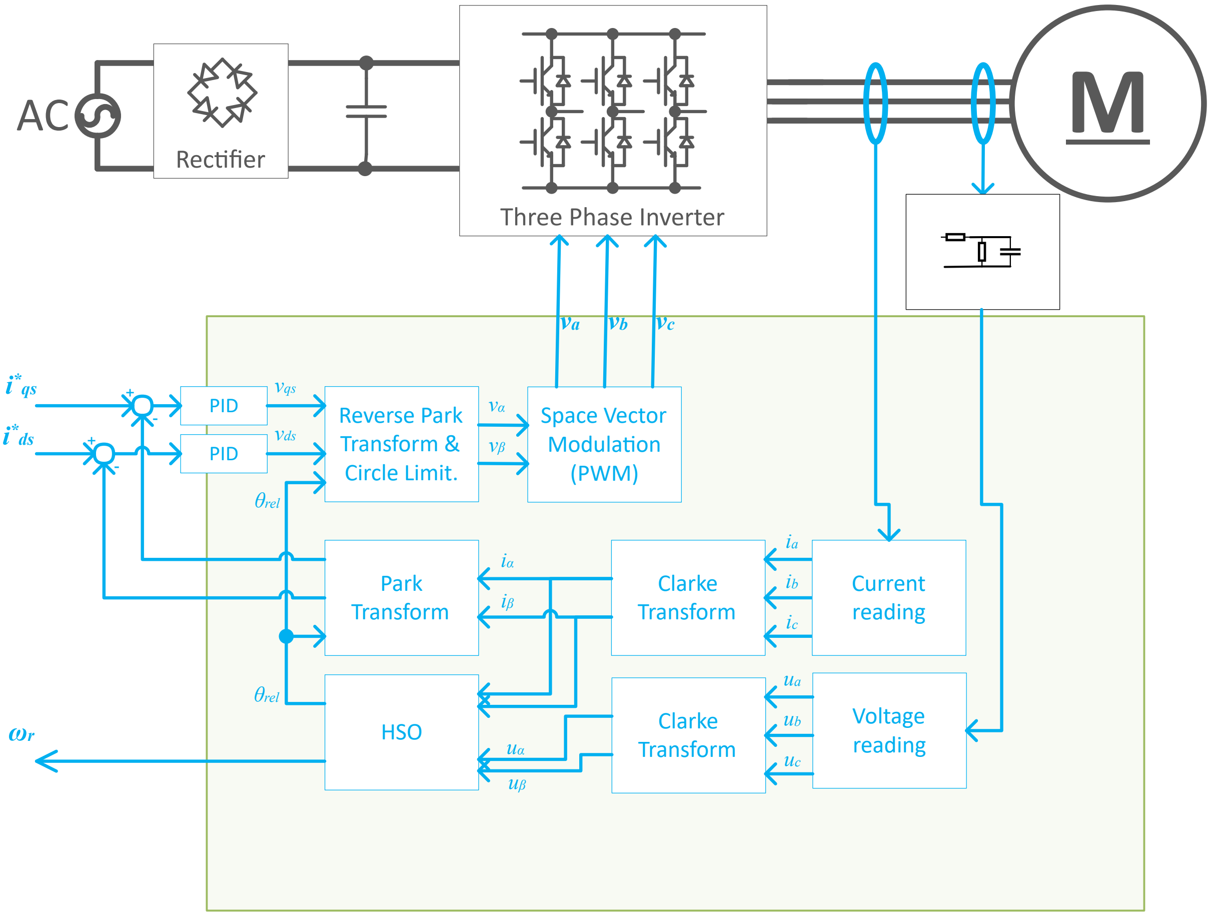
HSO startup guide

Introduction

High Sensitivity Observer (HSO) delivers an estimation of the angle and speed from measured phase currents and voltages, based on resistance Rs and inductance Ls



Back-emf e (in stationary coordinates: $e_{\alpha\beta}$) can be calculated from measured terminal voltages u_s and motor currents i_s with knowledge of stator resistance R_s and inductance L_s using (3). Besides valid values for R_s and L_s, pure differentiation and perfect voltage and current signals are needed in this case. Equation (2) implies that a correct value of e results to get a correct estimation of magnet ψ_{mag}.

1. $u_s(t) = U_{LR}(t) + \frac{d\psi_{mag}}{dt}$
2. $e(t) = \frac{d\psi_{mag}}{dt}$
3. $e(t) = u_s(t) - U_{LR}(t)$, where $U_{LR}(t)$ is the combined voltage across the inductance and resistance.

Back-emf is then integrated to get $\psi_{\alpha\beta}$ which will be used to estimate the angle thanks to G4 CORDIC peripheral. Speed is computed as the derivative of the angle.

As described above, very good voltage and current measurements are required. It is, therefore, strongly advised

- for current sensing:

to use Kelvin connection in case of three shunt topology and to use an analog RC filter to reject high frequency noise. Recommended cut-off frequency is 300KHz.

- for voltage sensing:

use an analog RC filter with recommended cut-off frequency of 500Hz.

HSO requires at least 2 ADCs to sense both currents and voltages, 3 ADCs is highly recommended.

Please refer to https://www.st.com/en/evaluation-tools/steval-lvlp01.html and https://www.st.com/en/evaluation-tools/b-g473e-zest1s.html, this pack is designed to use HSO/ZeST.

Control mode

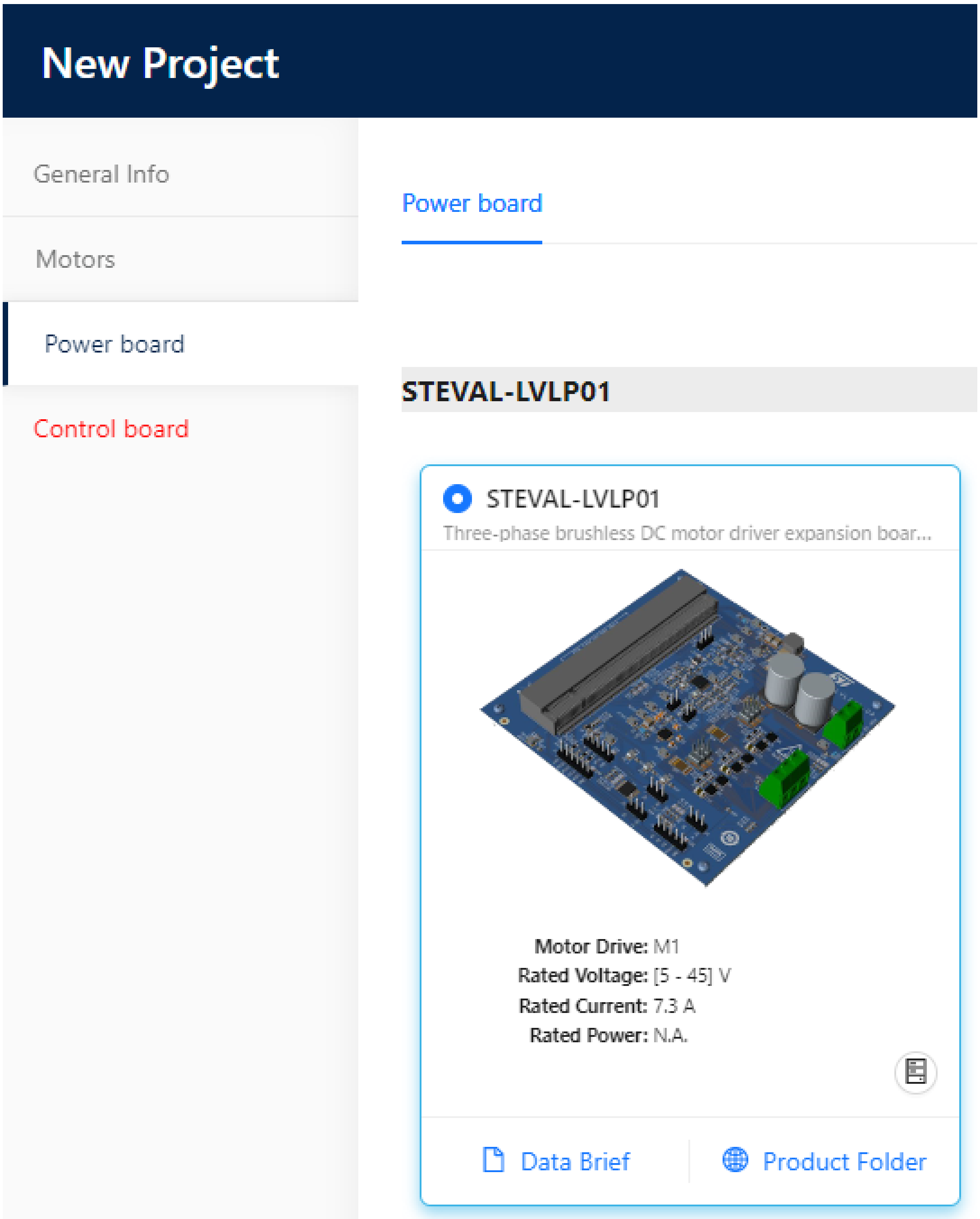
HSO firmware offers several control modes:

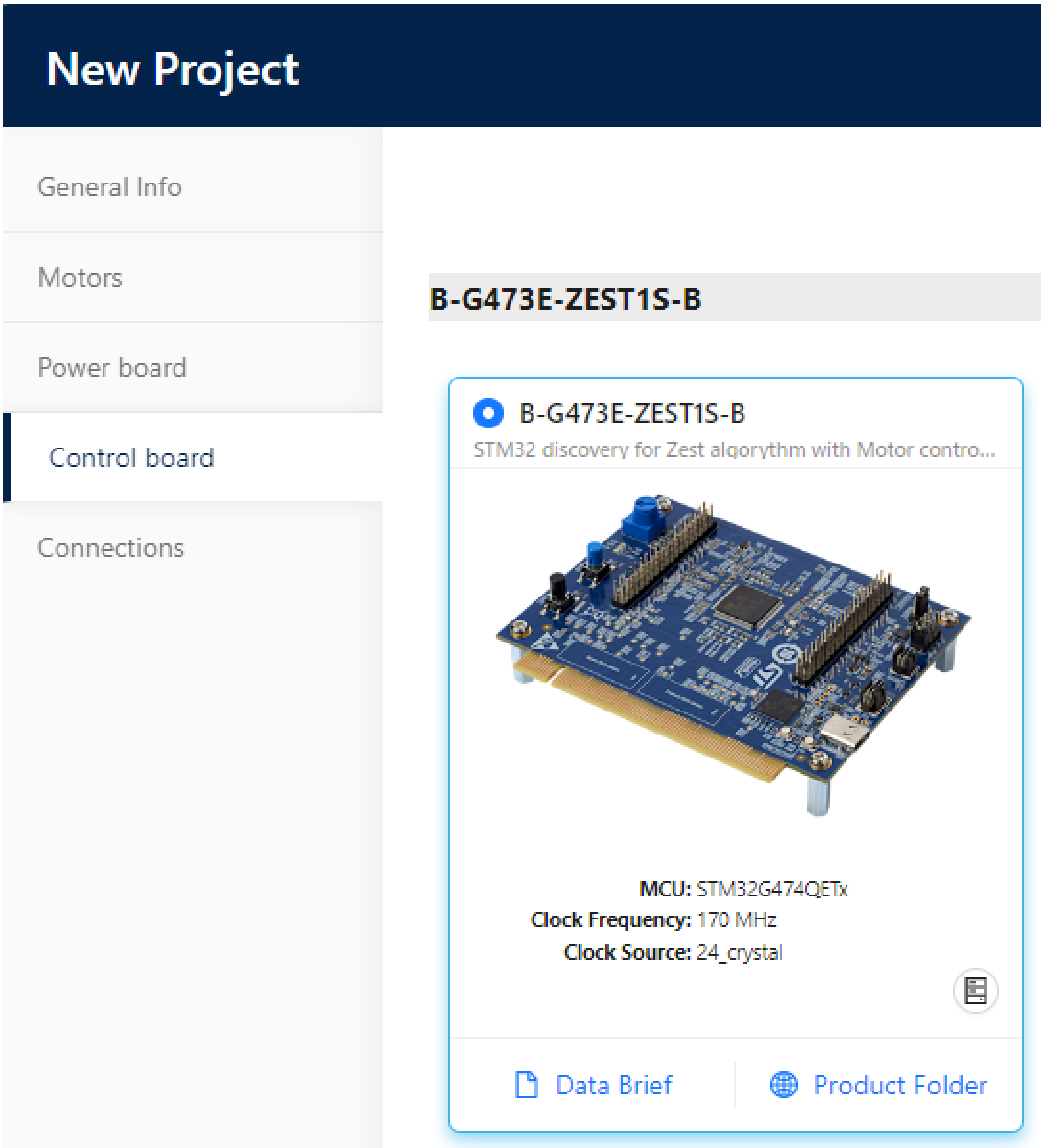
- `Observing` mode: in this mode Timer channels and Main Output (MOE) are disabled to open the transistors. The motor is then free to move and HSO is able to track it by reading the phase voltages. This mode is useful for flying start.
- `Open loop duty`: in this mode the firmware forces a rotating magnetic field by generating phase voltage without any current control. this mode is useful to debug a new hardware.
- `Open loop current`: in this mode the firmware forces a rotating magnetic field by generating phase voltage and current control. this mode is useful to debug a new hardware.
- `Closed loop speed`: in this mode the motor is tracked and the speed is controlled.
- `Closed loop torque`:in this mode the motor is tracked and the torque is controlled.
- `Profiling`: This mode is needed to use the motor profiler and only relevant if the project is created with motor profiler option selected.
- `Shorted`: in this mode the low sides are shorted.

Project creation

Board selection

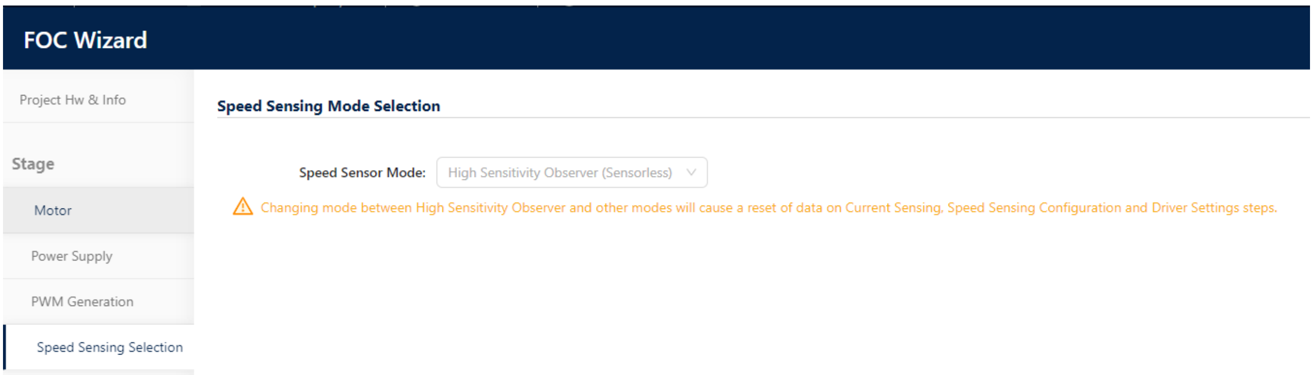
As HSO requires phase voltages measurements, only the boards that support them can be used to create a project with HSO. Currently, STEVAL-LVLP01 + B-G473-ZEST1S-B is the only official ST pack that supports it.


New Project
General Info
Motors
Power board
Control board
Power board
STEVAL-LVLP01
STEVAL-LVLP01
Three-phase brushless DC motor driver expansion boar...
Motor Drive: M1
Rated Voltage: [5 - 45] V
Rated Current: 7.3 A
Rated Power: N.A.
Data Brief
Product Folder

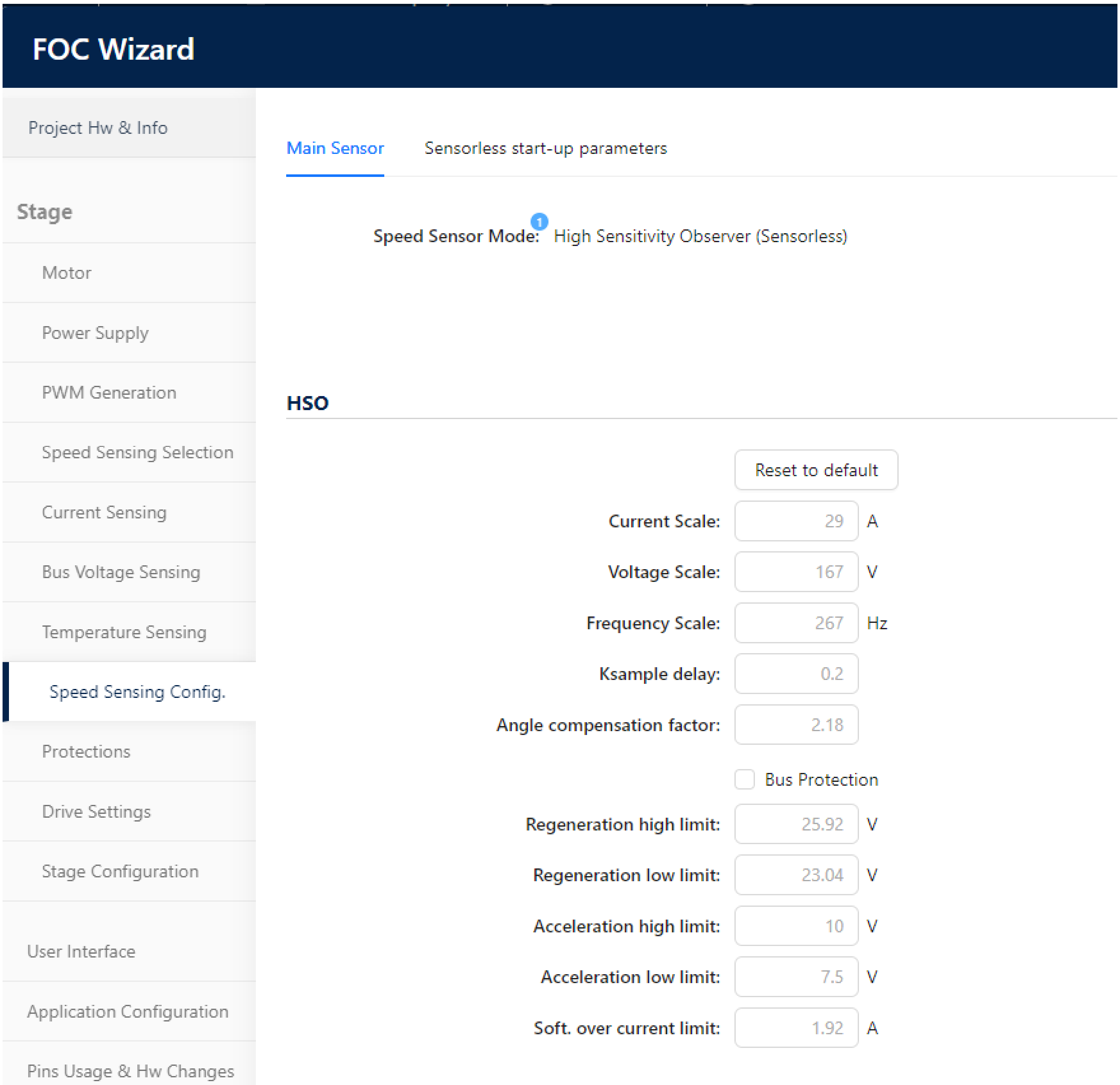


Speed sensing selection and configuration

If the setup supports the phase voltage sensing, MC Workbench should propose by default the HSO for speed sensing. User can ensure it is the case in `Speed sensing selection` tab.



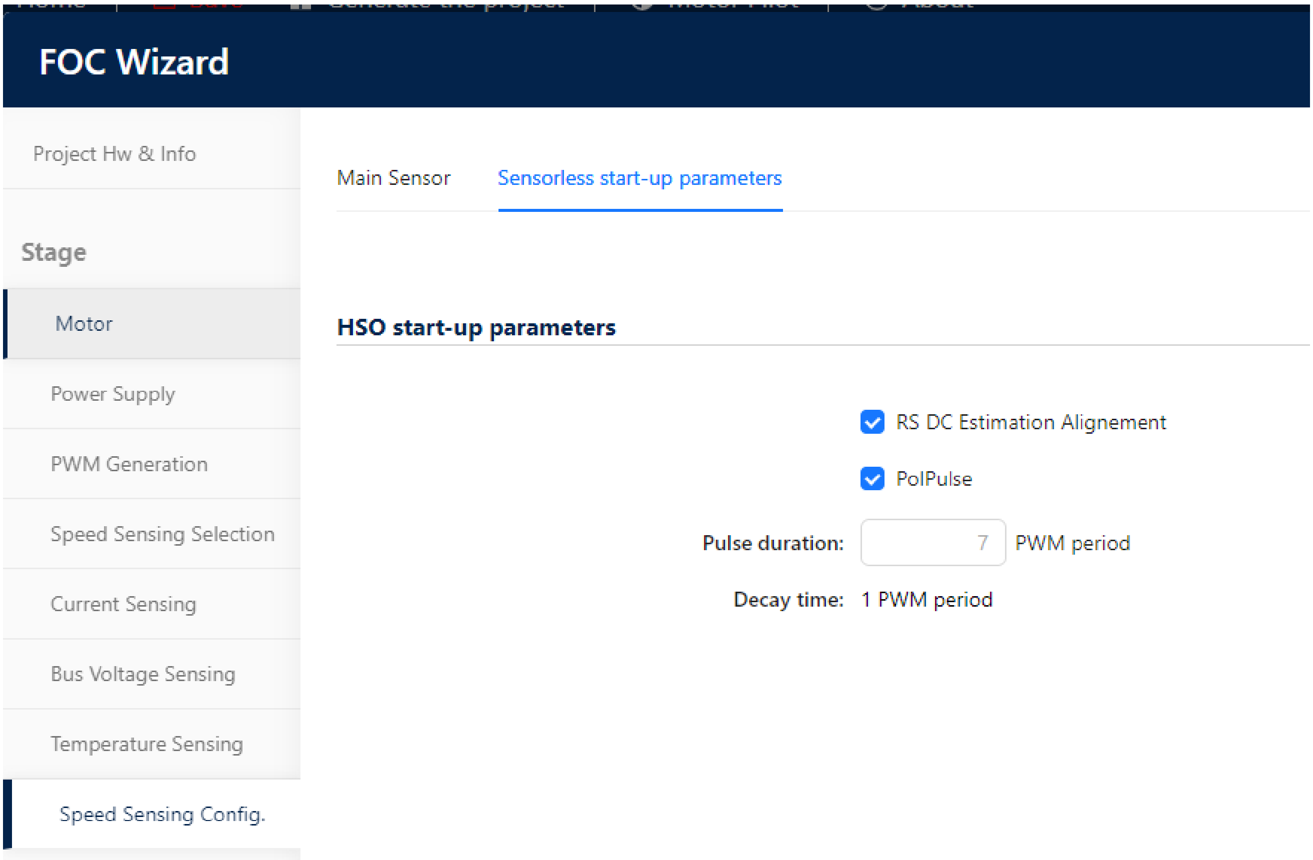
The configuration of the HSO can be done in the `speed sensing Config.` tab.



All this parameters above are preset by MC Workbench and can be fine tuned if needed.

Parameters	Description	Guidance
voltage scale	Calculation scale must be able to fit any measured/controlled voltage. optimal would be just large enough to fit to get best resolution.	This value is dependent on the maximum voltage your design can sense. This can be set to twice the voltage sensing range to start. If more resolution is needed, it can be reduced, but never below the maximum voltage that can be measured based on the hardware design.
current scale	Calculation scale must be able to fit any measured/controlled current. optimal would be just large enough to fit to get best resolution.	This value should be twice the maximum current that is measurable in the design.
frequency scale	Calculation scale must be able to fit any measured/controlled electrical frequency.	Optimal would be just large enough to fit to get best resolution.
Ksample delay	Impedance estimation is only correct at high frequencies when there is no timing difference between voltages and currents. To mitigate such a difference the parameter is to choose the 'weight' of the effective voltage sample at some point between the last and the previous sample	Refer to your configured ADC sample order. Set this to the delay based on the time between current sample and voltage sample. Units of this are in scale of duty cycle. E.g. 10kHz PWM (100us) value of 0.25 -> 25us.
angle compensation factor	Delay compensation factor which value is hardware and software dependent. It is used to compensate the delay between the Park angle (angle at the moment of ADC measurements) and the inverse Park angle (angle at the moment the PWM becomes effective)	Monitor the DutyCycleD and DutyCycleQ in MC Pilot. With the motor unloaded in speed control mode command a low speed to the motor 20Hz to 50Hz. Sweep the speed up to 80% max. speed while monitoring DutyCycleD and DutyCycleQ. If DutyCycleD is larger than 5% of DutyCycleQ further tuning is required. Increase delay compensation factor and repeat speed sweep until a value is reached to achieve DutyCyleD 5% less than DutyCycleQ.
Soft. overcurrent limit	Limit used by the firmware to perform a software protection against over current	

Note: by default the DC bus protection is disabled, this feature can be useful during *the* development phase but not intended to be used in *the* final product. Its description can be found in the MCSDK documentation/Components/Speed & torque control for HSO.



HSO proposes two methods for sensor-less startup:

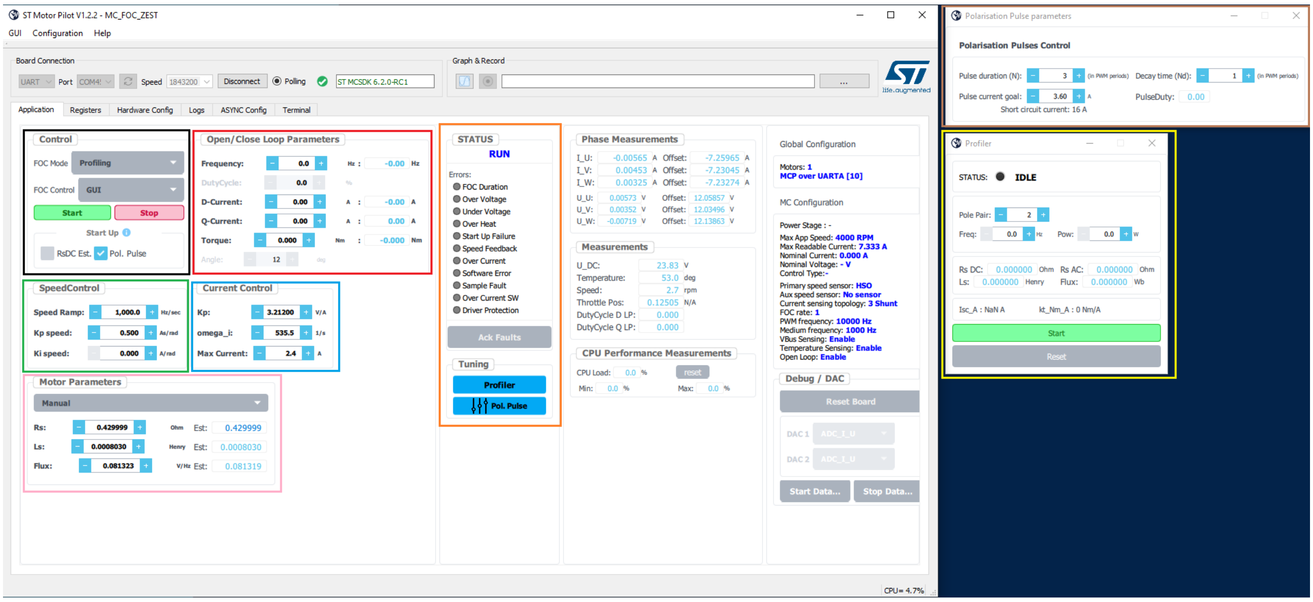
- RS DC Estimation and alignment which basically performs a rotor alignment
- PolPulse which detects the initial rotor position by generating current pulses. PolPulse parameters are preset by MC Workbench according to the motor parameters and their tuning can be found in the MCSDK documentation/User manual/Speed & position feedback sensorless algorithms or below.

  IMPORTANT: current pulses amplitude must be large enough to reach the inductance saturation level which is needed to get a correct estimation of the rotor position. This constraint implies that this method may be not suitable if the current level to saturate the inductance is higher than hardware and/or software over current limit of the system. In that case, RS DC Estimation and alignment should be used.

MC Pilot GUI

GUI Description

MC pilot supports HSO projects with the following interface:



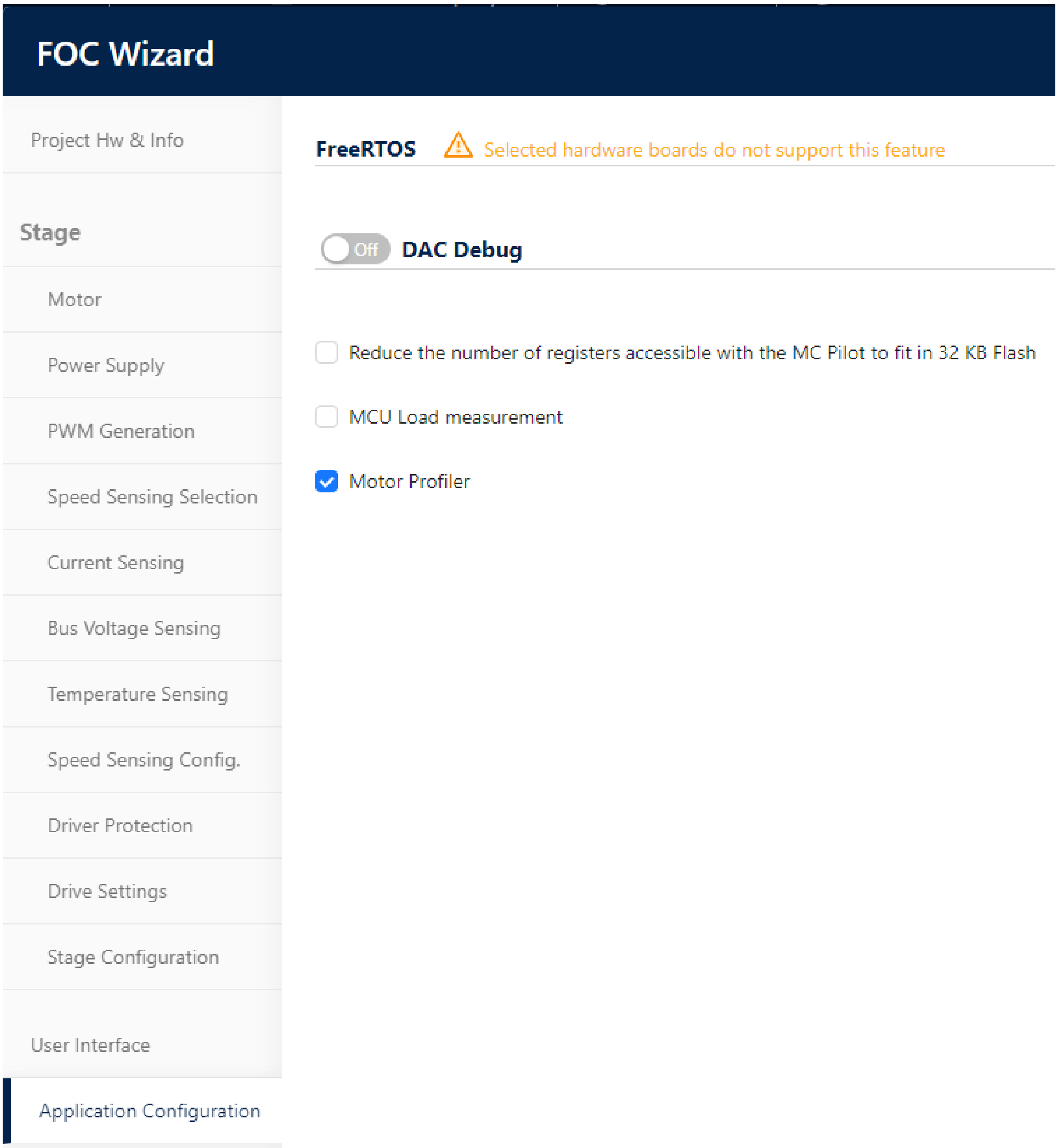
The GUI is divided in different blocks:

- `Control` block is used to configure the type control (black rectangle in the picture above):
  - ***FOC mode***: user can choose between observing, Open loop duty, Open loop current, closed loop speed, closed loop torque, profiling, shorted. these modes are described above.
  - ***FOC Control***: the target (speed or torque) is set via the GUI or via potentiometer. Only GUI is supported.
  - ***Start*** button: starts motor in the selected ***FOC mode***.
  - ***Stop*** button: stops motor.
  - ***Pol.Pulse*** checkbox: enables Pol.Pulse feature to startup the motor.
  - ***Rs.DC*** checbox: enables Rs.DC feature to startup the motor.

  Note: ***Pol.Pulse*** and ***Rs.DC*** mutually exclusive.

- `Open/Close loop parameters` block is used to set the target according to the type of selected FOC mode (red rectangle in the picture above).
  - ***Frequency***: motor electrical frequency target. Formula to convert motor mechanical speed into electrical frequency is $F_{(Hz)} = (Speed_{(rpm)}/60) * pole\ pair\ number$.
  - ***duty cycle***: PWM duty cycle modulation level in percent. only relevant in Open loop duty mode.
  - ***D-current***: Id current target.
  - ***Q-current***: Iq current target, only relevant in torque mode.
  - ***Torque***: torque target.
  - ***Angle***: rotor angle, editable only in open loop duty/current mode.
- `Speed control` Speed control block allows speed control tuning by setting new speed PI gains and a different speed ramp (Hz/s) (green rectangle in the picture above)
- `Current control` block allows to tune the current control by setting the current PI gains and the maximum current available for the application. This value is clamped with the minimum value between the motor maximum current and board maximum current. (blue rectangle in the picture above)
- `Motor parameters` lists the motor flux, resistance and inductance. These parameters can be changed at run time (pink rectangle in the picture above)
- `Status` lists the state machine status and the error flags (orange rectangle in the picture above). Errors and status descriptions can be found in the MCSDK documentation/User manual/MC State Machine, Commands & Faults.

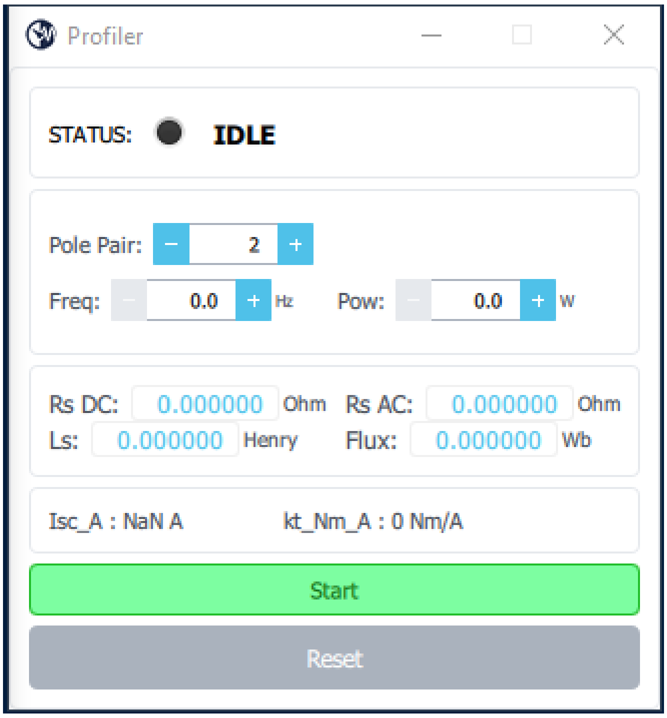
- `tuning` contains two buttons:
  - [Profiler]() opens a windows to configure motor profiler feature (yellow rectangle in the picture above). **Important**: Profiler button appears only if the motor profiler has been enabled in the MC Workbench project as shown below.



  - [PolPulse]() opens a window to tune PolPulse feature (brown rectangle in the picture above).

Motor Profiler GUI

HSO project provides a motor profiling feature which allows to estimate motor resistance, inductance and flux. The profiling result is automatically transferred to firmware at run-time which allows user to control the motor directly after profiling procedure.



Before starting the profiling, user must set:

- motor pole pair number.
- target motor electrical frequency. Default value is 50Hz,this value can be tuned according to the load of the motor. The speed should be high enough to make the motor spin in order to estimate the flux. Usually a value between 20Hz and 50Hz is sufficient.
- Power level. Default value is 3W, this value can be tuned according to board maximum current and motor load. 1% of the rated power is usually sufficient.

Then start the profiler and wait for completion. To execute another profiling process, user needs to press reset button and then start again.

Note: when profiling is complete, the frequency scaling may need to be adjusted according to the maximum speed of the profiled motor. It advised to set a value two times greater than the estimated maximum electrical speed returned by the profiler. This modification can be done in the `Hardware Config` panel.

Status:

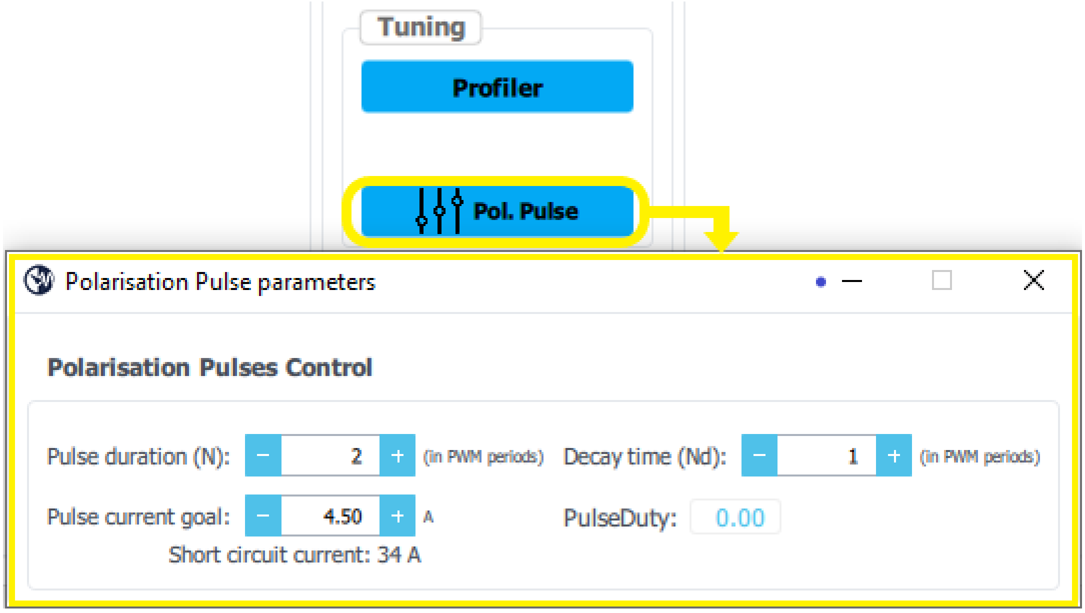
- `IDLE`: profiler is in idle state and wait for a *start* command.
- `DC_AC_CHECK`: profiler is measuring motor resistance and inductance.
- `FLUX_ESTIMATION`: profiler is measuring the motor flux.
- `COMPLETED`: motor profiling is complete. User must press *reset* button before starting a new profiling process.
- `ERROR_AC_DC_CHECK`: error occurs when DCAC profiler step failed, when measured resistance or inductance are out of expected ranges. User should increase the power level and press *reset* button to clear the error.
- `ERROR_FLUX_ESTIMATION`: error occurs when motor fails to spin during profiling. Usually this error occurs because the power target is not high enough to make the motor spin. User should increase the power level and press *reset* button to clear the error.

Important:

- Profiling result is not saved into a file, user needs therefore to manually port them in the project.
- When motor profiler is enabled in ST MC Workbench project, the frequency scaling factor is forced 1000Hz. It is advised, therefore, to not drive the motor at an electrical frequency greater than 500Hz after profiling. If the user wants to go beyond, assuming the motor is able to do it, he will needs to update its ST MC Workbench project or use the Hardware Config panel to update the parameters.
- PI speed gain are not computed, user must tune them after profiling.
- HSO needs accurate Rs, Ls and Flux to work properly, it is therefore strongly advised to profile again motors that have been previously profiled by the legacy profiler.

PolPulse GUI

HSO can make use of the PolPulse algorithm to determine the position of the rotor at startup. Indeed, the Polarity detection through Pulsing feature improves significantly the startup time of the motor compared to a basic alignment. It also avoids any movement of the rotor before startup. MC Pilot GUI allows to find the correct settings to configure the pulses generation to help the algorithm to determine the actual angle of a stand-still machine.



• **Pulse duration** : number of PWM periods to reach the current goal. This value will be used to compute a duty cycle (**PulseDuty**) for the PWM generation and should be checked carefully.

• **Decay time** : number of PWM periods for the current to decay. It depends on the motor inductance, it can be tuned by connecting current probes on the motor phases and checking the pulses on the oscilloscope (an example is shown on the picture below). When decay is finished the currents level should be zero, if not increase the decay time.

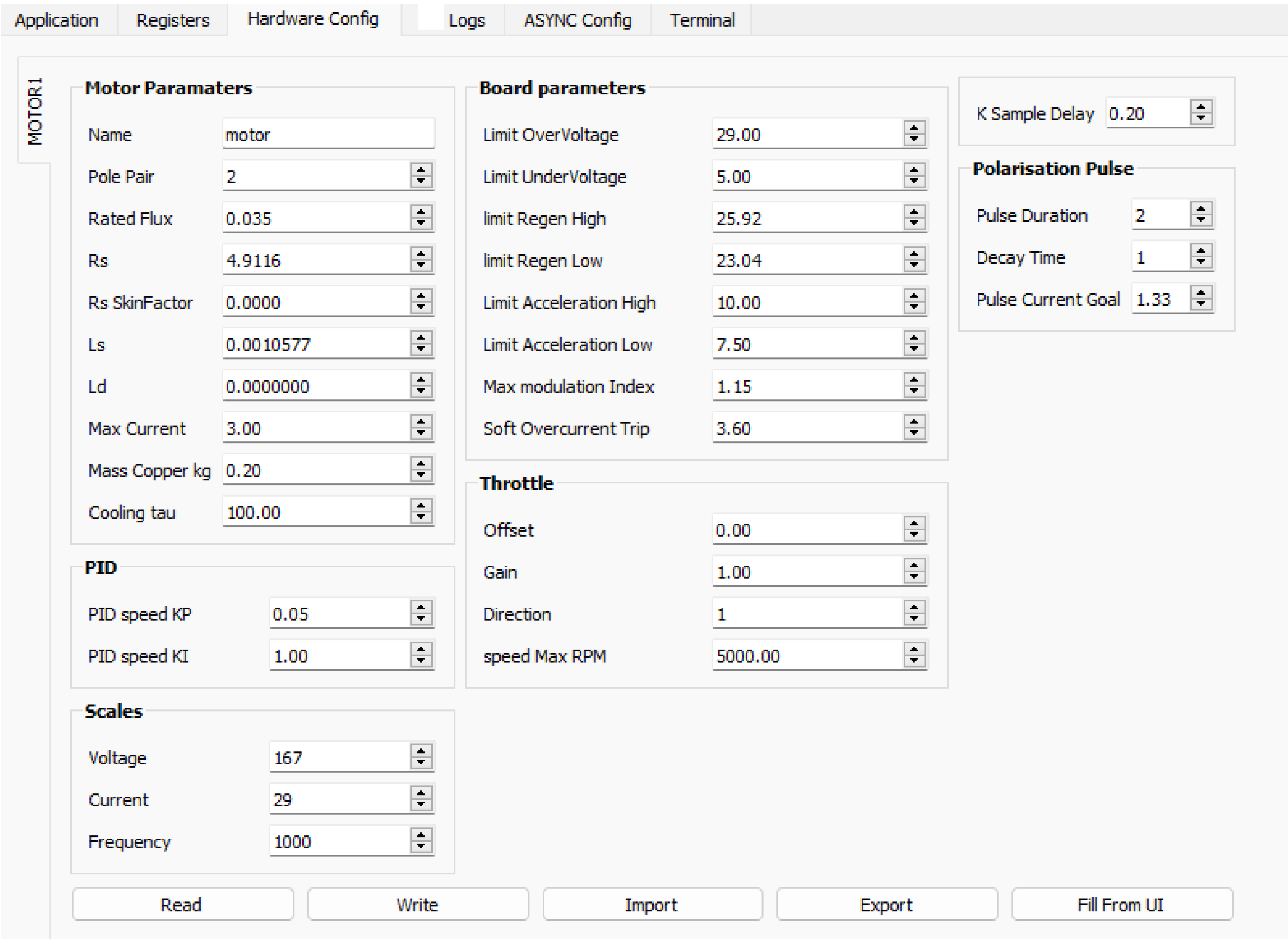
• **Pulse current goal**: level of pulse current expected. By default, it is set to $\frac{1}{4}$ of the short circuit current and limited to the board soft overcurrent trip value. Currently, pulse current goal cannot be modified in the ST MC Workbench. If user wants to change it without using MC Pilot, it can be done by modifying the `PulseCurrentGoal_A` value (expressed in Ampere) which can be found in the routine `MC_POLPULSE_init()` in the file mc_polpulse.c.

• **PulseDuty**: Computed duty cycle according to the **pulse duration** and **Pulse current goal** configured by the user. This value is clamped between -0.98 and 0.98, therefore user shall configure **pulse duration** in order to get the highest possible absolute value strictly lower than 0.98.

In this example the pulse duration (N) is 2, the decay time (Nd) is 2. The current pulses is plot in channel 6.

Hardware Config

`Hardware Config` panel allows to modify a set of parameters stored in the flash without the need of recompiling the project. This can be useful after a profiling to adapt scaling factors or when swapping to an already profiled motor.



- **Read**: reads the set of parameters from the flash.
- **Write**: writes the set of parameters into the flash.

- **Import**: import the set of parameters from file.
- **Export**: export the set of parameters to a file.
- **Fill from UI**: fills the following fields with the ones extracted from the UI: `Rated Flux`, `Rs`, `Rs skinfactor`, `ls`, `PID Speed Kp`, `PID Speed Ki`, `K sample delay`, `Pulse duration`, `Decay Time` and `Pulse Current Goal`.

Note: when the user writes the new set of parameters into the flash, a MCU reset is needed to reinitialize the firmware with the new parameters.

GUI usage

Usual flow when using a new motor:

- Create a project with **Open loop** and **Motor Profiler** activated, and flash it.
- Select `profiling` in the drop box `FOC mode`, click on `Start` button and wait for the `status` to be in RUN as shown in the Motor Pilot GUI below.

- Profiler the motor. At he end of the profiling, flux,resistance and inductance will be estimated. **Friction and inertia are not estimated by the profiler therefore PI speed gain must be tuned by the user**.
- At the end of a profiling, PolPulse **must** be tuned. Profiler only estimates the pulse current goal ($I = \frac{1}{4} . \frac{\psi}{L}$) which must considered as a starting point. If PolPulse is not correctly tuned, it may generate over current errors at startup or motor unable to spin.
- Click on `Stop` button to stop the motor.
- Select a mode in `FOC mode` list.
- Set the desired `speed` or `torque` target.
- Click on `RsDC est.` or `PolPulse` checkbox for the startup.
- Click on `Start` button to start the motor.
- Click on `Stop` button to stop the motor.

Usual flow when using an already profiled motor (profiler and open loops disabled in the WB project):

- Create a project and flash it.
- Select a mode in `FOC mode` list.
- Set the desired `speed` or `torque` target.
- Click on `RsDC est.` or `PolPulse` checkbox for the startup method.
- Click on `Start` button to start the motor.
- Click on `Stop` button to stop the motor.